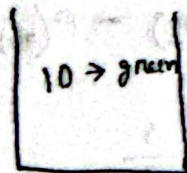


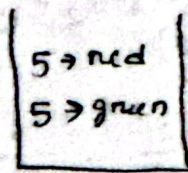
TOPIC NAME :

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30/8/25

Decision Tree $\rightarrow$ classification

entropy = 0



Entropy = 1

$$\text{Entropy} = -P_i \log_2 P_i$$

 $i = \text{unique value}$

step :

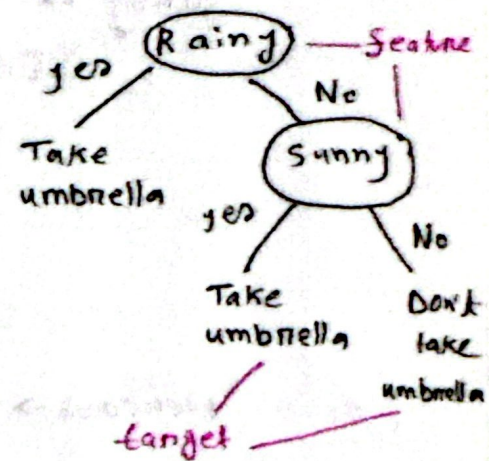
i) Target $\rightarrow$ entropy

ii) For each feature,

find entropy and information gain

iii) Sort $\rightarrow$ Descending Order

iv) create the tree



$$\text{Target(entropy)} = -P_{\text{yes}} \log_2 P_{\text{yes}} - P_{\text{No}} \log_2 P_{\text{No}}$$

$$= -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14}$$

$$= 0.94$$

Day	Feature				Target
	Outlook	Temp	Humidity	Wind	Play Tennis
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

TOPIC NAME : _____

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Outlook

Sunny →

$$\begin{aligned} \text{Entropy} &= -P(\text{yes}|\text{sunny}) \log_2 P(\text{yes}|\text{sunny}) - P(\text{no}|\text{sunny}) \log_2 P(\text{no}|\text{sunny}) \\ &= -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} \\ &= 0.97 \end{aligned}$$

Overcast →

$$\begin{aligned} \text{Entropy} &= -P(\text{yes}|\text{o}) \log_2 P(\text{yes}|\text{o}) - P(\text{no}|\text{o}) \log_2 P(\text{no}|\text{o}) \\ &= -\frac{4}{4} \log_2 \frac{4}{4} - \frac{0}{4} = 0 \end{aligned}$$

Rainy →

$$\begin{aligned} \text{Entropy} &= -P(\text{yes}|\text{r}) \log_2 P(\text{yes}|\text{r}) - P(\text{no}|\text{r}) \log_2 P(\text{no}|\text{r}) \\ &= -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} \\ &= 0.97 \end{aligned}$$

→ Information gain of outlook

$$\begin{aligned} \text{IG (outlook)} &= \text{Target Entropy} - (\text{weighted sum of entropy of features}) \\ &= 0.94 - \left(0.97 \times \frac{5}{14} + 0 \times \frac{4}{14} + 0.97 \times \frac{5}{14} \right) \\ &= 0.247 \end{aligned}$$

TOPIC NAME : _____

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Temperature

Hot $\rightarrow$

$$\begin{aligned} \text{Entropy} &= -P(\text{yes}|H) \log_2 P(\text{yes}|H) - P(\text{No}|H) \log_2 P(\text{No}|H) \\ &= -\frac{3}{4} \log_2 \frac{3}{4} - \frac{2}{4} \log_2 \frac{2}{4} \\ &= 1 \end{aligned}$$

Mild $\rightarrow$

$$\begin{aligned} \text{Entropy} &= -P(\text{yes}|M) \log_2 P(\text{yes}|M) - P(\text{No}|M) \log_2 P(\text{No}|M) \\ &= -4/6 \log_2 4/6 - \frac{2}{6} \log_2 \frac{2}{6} \\ &= 0.91 \end{aligned}$$

Cool $\rightarrow$

$$\begin{aligned} \text{Entropy} &= -P(\text{yes}|C) \log_2 P(\text{yes}|C) - P(\text{No}|C) \log_2 P(\text{No}|C) \\ &= -\frac{3}{4} \log_2 \frac{3}{4} - \frac{1}{4} \log_2 \frac{1}{4} \\ &= 0.81 \end{aligned}$$

$$\begin{aligned} IG(\text{temperature}) &= 0.94 - \left(1 \times \frac{4}{14} + 0.918 \times \frac{6}{14} + 0.81 \times \frac{4}{14} \right) \\ &= 0.029 \end{aligned}$$

GOOD LUCK™

TOPIC NAME : _____

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Humidity

High →

$$\text{Entropy} = - \frac{3}{7} \log_2 \frac{3}{7} - \frac{4}{7} \log_2 \frac{4}{7} = 0.985$$

Normal →

$$\text{Entropy} = - \frac{6}{7} \log_2 \frac{6}{7} - \frac{1}{7} \log_2 \frac{1}{7} = 0.591$$

$$\text{IG (Humidity)} = 0.94 - \left(\frac{7}{14} \times 0.985 + \frac{7}{14} \times 0.591 \right)$$

$$= 0.152$$

Wind

Weak →

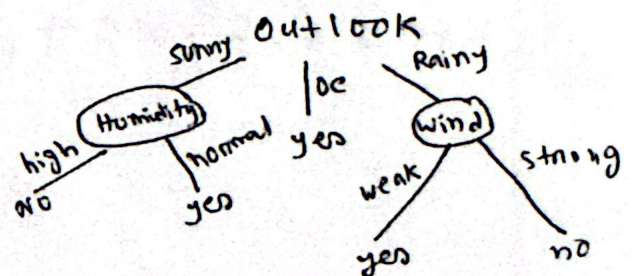
$$\text{Entropy} = - \frac{6}{8} \log_2 \frac{6}{8} - \frac{2}{8} \log_2 \frac{2}{8} = 0.811$$

Strong →

$$\text{Entropy} = 1$$

$$\text{IG (wind)} = 0.94 - \left(0.811 \times \frac{8}{14} + 1 \times \frac{6}{14} \right)$$

$$= 0.048$$



Day	outlook	Temp	humidity	wind	play tennis
1	sunny	Hot	High	weak	no
2	Sunny	Hot	High	strong	no
8	sunny	Mild	High	weak	no
9	sunny	cool	normal	weak	yes
11	sunny	Mild	normal	strong	yes

$$\text{Entropy target} = -P_{\text{yes}} \log_2 P_{\text{yes}} - P_{\text{no}} \log_2 P_{\text{no}}$$

$$= -\frac{2}{5} \log_2 \frac{2}{5} - \frac{3}{5} \log_2 \frac{3}{5} = 0.97$$

Temp

$$\text{Entropy (Hot)} = 0$$

$$\text{Entropy (cool)} = 0$$

$$\text{Entropy (Mild)} = 1$$

$$IG(\text{Temp}) = 0.97 - \left(1 \times \frac{2}{5}\right) = 0.57$$

Humidity

$$IG = 0.97$$

Wind

$$\text{Entropy (weak)} = -\frac{1}{3} \log_2 \frac{1}{3} - \frac{2}{3} \log_2 \frac{2}{3} = 0.918$$

$$\text{Entropy (strong)} = 1$$

$$IG(\text{wind}) = 0.97 - \left(0.918 \times \frac{3}{5} + 1 \times \frac{2}{5}\right) = 0.0192$$

TOPIC NAME : _____

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Day	outlook	Temp	Humidity	Wind	Play Tennis
4	Rain	mild	high	weak	yes
5	Rain	cool	normal	weak	yes
6	Rain	cool	normal	strong	no
10	Rain	mild	normal	weak	yes
14	Rain	mild	high	strong	no

$$\text{Entropy (target)} = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.97$$

Temp

$$\text{Entropy (cool)} = 1$$

$$\text{Entropy (mild)} = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3}$$

$$= 0.918$$

$$IG = 0.97 - \left(1 \times \frac{2}{5} + 0.918 \times \frac{3}{5} \right) = 0.019$$

Humidity

$$\text{Entropy (High)} = 1$$

$$\text{Entropy (low)} = -\frac{2}{3} \log_2 \frac{2}{3} - \frac{1}{3} \log_2 \frac{1}{3} = 0.918$$

$$IG (\text{Humidity}) = 0.97 - \left(1 \times \frac{2}{5} + 0.918 \times \frac{3}{5} \right) = 0.019$$

$$IG (\text{wind}) = 0.97$$